

ORIGINAL RESEARCH

Evaluation and calibration of low-cost off-the-shelf particulate matter sensors using machine learning techniques

Mohammad Ghamari¹  | Hamid Kamangir² | Keyvan Arezoo³ | Khalil Alipour³

¹Department of Electrical and Computer Engineering, Kettering University, Flint, Michigan, USA

²University of California-Davis, Biological and Agricultural Engineering, Davis, California, USA

³Department of Mechatronics Engineering, University of Tehran, Tehran, Iran

Correspondence

Mohammad Ghamari, Department of Electrical and Computer Engineering, Kettering University, Flint, MI, 48504, USA.
Email: mghamari@kettering.edu

Abstract

The use of inexpensive, lightweight, and portable particulate matter (PM) sensors is increasingly becoming popular in air quality monitoring applications. As an example, these low-cost sensors can be used in surface or underground coal mines for monitoring of inhalable dust, and monitoring of inhalable particles in real-time can be beneficial as it can possibly assist in preventing coal mine related respiratory diseases such as black lung disease. However, commercially available PM sensors are not inherently calibrated, and as a result, they have vague and unclear measurement accuracy. Therefore, they must initially be evaluated and compared with standardised instruments to be ready to be deployed in the fields. In this study, three different types of inexpensive, light-scattering-based widely available PM sensors (Shinyei PPD42NS, Sharp GP2Y1010AU0F, and Laser SEN0177) are evaluated and calibrated with reference instruments. PM sensors are compared with reference instruments in a controlled environment. The calibration is done by means of different machine learning techniques. The results demonstrate that the calibrated response obtained by fusion of sensors has a higher accuracy in comparison to the calibrated response of each individual sensor.

KEYWORDS

air quality, low-cost sensors, machine learning, particulate matter sensor, sensor calibration, sensor fusion

1 | INTRODUCTION

Nowadays, the air pollution is one of the main challenges of large and industrialised cities of the world [1]. The air pollution has increased the risk of death and disease all over the world [2]. Hence, the negative and serious economic, social, and mental effects of air pollutants should be addressed [3]. It is estimated that air pollution is the cause of 1.4% of the total death in the world [4]. This problem has been examined in detail in the US [5], UK [6], France [7], and many countries in the European Union [8] and in this regard, many successful efforts have been done to measure the amount of density of pollutant in the air [4, 9, 10]. Among various pollutant, the Particulate Matter (PM) is very important. Particulate matter can have both natural and synthetic origins [11]. This particle pollution is a heterogeneous mixture of solid and liquid particles that are suspended in air and are different with each

other from size and chemical contents [12]. In a study that was performed at Zhejiang province of China in 2013 and 2017, the mortality rate related to PM₁, PM_{2.5}, and PM₁₀ were estimated 2.39%, 2.53% and 3.08% respectively [13]. More specifically, particles that have diameters smaller than 10 micrometres have adverse health effects since they can penetrate to the lungs [13]. In addition, some of the particles can enter the blood flow and therefore can lead to sudden heart attack, reduction of lung performance, non-uniform heart pulse rate, increase in severity of asthma, and sudden death in people who suffer from heart and respiratory diseases [13]. In addition to previously mentioned adverse health related effects, these particles can lead to the reduction of horizontal visibility and in some cases vertical visibility in aerial, ground and naval transportations [14]. In addition to existing concerns associated with air pollution in urban environments, particle pollution (more specifically respirable

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coal dust) in coal mines has also been a controversial problem. Overexposure to coal dust can be harmful and thus may lead to lung-related diseases such as Coal Workers' Pneumoconiosis (CWP) commonly known as black lung disease [15, 16]. CWP is an occupational lung disease caused by inhalation of coal dust which can cause possible damage to the lungs' alveoli [17]. The advanced form of CWP is known as progressive massive fibrosis [18] which can generally be deadly [19]. To better protect the coal workers from getting overexposed to respirable dust, recently the Mine Safety and Health Administration (MSHA) has suggested a new law which includes reduction of the concentration of the respirable dust from 2 mg of dust per cubic metre of air (mg/m^3) to 1.5 mg/m^3 [20]. Therefore, the respirable coal dust concentration based on new law needs to be reevaluated in mine perimeters to comply with the suggested rule. Nowadays, there are various tools and methods to measure the size and concentration of PM. These tools are based on three methods of measurement including microbalance, optical and gravimetric [21]. In gravimetric method, the weight and concentration of PM collected at filters are measured before and after sampling time [22]. In optical method, concentration and size of PM are measured based on scattering, receiving, and not receiving of the light [22]. Microbalance-based tools use the change of resonance frequency to determine PM [23]. There are two main measuring tools which utilise microbalance principle: Tapered Element Oscillating Microbalance (TEOM) and Quartz Crystal Microbalance [23]. While data obtained from this method is very accurate and reliable, but they cannot instantly present the collected results, plus they cannot communicate in a smart network to exchange data in real-time to cover a wide geographical area. Thus, this method is not a preferred method of monitoring when real-time presentation of results is required. As a result, an alternative real-time monitoring and cost-efficient system is required to observe the air quality. Currently, many of the principal manufacturers of equipment offer the use of low-cost sensors, such as Shinyei PPD42NS, Sharp GP2Y1010AU0F and Laser SEN0177 to evaluate the concentration of PM. Yet, there are serious concerns regarding their precision and calibration [24]. Since the calibration of these low-cost sensors was performed in a controlled environment at manufacturer, their calibration is not precise once they are placed at a real environment. Therefore, they need to be recalibrated to provide accurate results. As yet, calibration of low-cost PM sensors has been researched extensively by many scientists. Many calibration techniques used are based on simple statistical methods such as linear regression. As an example, Ghamari et al. in [25] used a simple linear regression technique to demonstrate the relationship between the response of low-cost sensors with reference instruments. Statistical linear regression techniques have been used as a core method for calibration in many articles such as in refs. [26, 27]. Although, calibration techniques based on statistical methods are relatively accurate, new techniques based on Machine Learning (ML) algorithms can provide more accurate results. Until now, many calibration techniques have already been proposed by the use of ML

methods, from linear to Support Vector Regression (SVR). Maag et al. examined the effect of internal and external sources of error in calibration of cost-efficient sensors [28]. Castell et al. presented important results which support the use of ML methods to improve the accuracy of low-cost sensors [29]. Machine Learning techniques such as Multiple Linear Regression (MLR), SVR, Random Forest (RF) have been examined by Bigi et al. in ref. [30]. The authors in ref. [30] showed that the performance of nonlinear methods (SVR and RF) was better in comparison with linear methods. Another method to calibrate off-the-shelf sensors, which has been suggested by Spinelle et al., was Adaptive Neural Network [31]. It was shown this method could enhance the accuracy of calibration. Zimmerman et al. demonstrated that the performance of RF for calibration was better than MLR [32]. However, other approaches have also been suggested for calibration as alternatives to ML methods such as work that was performed in ref. [33], there still a deeper knowledge of models and parameters of the environment is required. The use of up to four sensors along with sensor fusion and MLR techniques has been used for calibration in ref. [34]. Authors in ref. [34] showed that the calibration results obtained from their proposed techniques led to better accuracy as compared to the accuracy of the best of four sensors [34]. Other techniques for sensor fusion have been focused on statistical methods for combination of data [35, 36]. To date, the performance of various low-cost PM sensors in terms of precision, soundness, noise, and detection limitation has been evaluated in laboratory and field conditions using a variety of data fusion methods. It is worth mentioning that the use of inexpensive sensors offers a cost-effective alternative to complete precision measurements. However, due to the low data quality obtained from these sensors, as mentioned earlier, the calibration of low-cost sensors has been studied in recent years to improve the quality of data. Limited studies investigated the possibility of improving data quality by using ML techniques. However, many of the ML approaches that have been used so far were mostly focused on evaluation of dust particles for remote sensing application [37, 38]. In some other research works, ML techniques and local sensors have been used to calibrate O_3 , NO_2 , and NO sensors to estimate air pollution [39–42]. Although using the ML models for the calibration of low-cost sensors can be useful and play a vital role in this field, most of the previous research works were focused on using only one of the ML approaches; in addition, one of the causes of poor performance in ML algorithms which occur because of overfitting problem has not been investigated properly [39, 43, 44]. To eliminate the effect of overfitting problem and to identify a more appropriate learning algorithm, authors in ref. [45] evaluated the performance of 9 ML regression algorithms for calibration of low-cost sensors which they were used for air-pollution evaluation. The aim of this research work is also to investigate the performance of different ML regression algorithms for calibration of low-cost PM sensors. Although several ML approaches that have been used in this research work, overlaps with the work that has been done in ref. [45], however, this research

further investigates two more additional ML algorithms such as Elastic Net and Stacking Regression in addition to the previously mentioned ML algorithms that have already been evaluated for calibration purposes and shown in ref. [45]. In addition, in contrast to the work that has been done in [45], in this research, authors used different set of data and the data also has been collected in a controlled environment by the authors exclusively. Besides, while the application of the research that has been conducted in ref. [45] is focused on evaluation and calibration of low-cost sensors in ambient conditions with the hope to minimise the harmful effects of air pollution, the application of this research is slightly different. The focus of this research is relied on evaluation of the quality of air in surface coal mines where different types of respirable dust particles such as coal dust that can be inhaled and cause serious health problems are involved. This paper has proposed a low-cost sensory system for evaluation of dust concentration via ML-based fusing algorithms. To this end, an innovative approach to improve the data quality of three inexpensive well-known PM sensors (Shinyei PPD42NS, Sharp GP2Y1010AU0F, and Laser SEN0177) is proposed for improving the accuracy of data. In this work, different sizes and types of particles (three different sizes of Arizona Road Dust (ARD) (A1 Ultrafine Test Dust, A2 Fine Test Dust, and A3 Medium Test Dust as well as one type of coal dust) was used in order to evaluate and compare the performance of these sensors. Various approaches are detailed that they have fused the sensors data to estimate their actual performance and further calibrate to improve their accuracy. To this end, eight different ML techniques and models are used to analyse and calibrate the characteristics of the sensors in laboratory conditions. The obtained results prove that the accuracy of results from a combination of sensors' data using ML techniques is improved compared to previous calibration that was performed on a single sensor. As a result, the contributions of the present study are listed as follows:

- A unique setup containing three different PM-types sensor is proposed for measuring the dust concentration.
- Eight ML approaches are compared to each other to calibrate the data obtained from the proposed sensory system.
- Results has shown that the accuracy of sensors' calibration using fusion algorithms, is improved compared to previous calibration that was performed on a single sensor.
- While the proposed calibration techniques based on ML approaches in this research can be used in different environmental conditions as data needs to be trained only once, previous simple linear regression models were limited to specific environments and to be used in other environments, data had to recalibrated.
- The proposed calibration techniques based on ML approaches is the first step in advancing this research towards the field of embedded ML where embedded systems incorporating low-cost processors can use the existing trained data in cloud to independently make decisions and predictions.

2 | MATERIALS AND METHODS

2.1 | Low-cost particulate matter sensors

In this study, three different types of off-the-shelf PM sensors (Shinyei PPD42NS, Sharp GP2Y1010AU0F, Laser SEN0177), which operate based on light-scattering techniques have been selected for performance evaluation. Each PM sensor proposed to be used here include a light source (usually an IR LED), a light-detecting component (e.g. photodiode), and an optical lens. The light source emits a light beam that is passed through the dust particles, and the photodetector is utilised to collect the dispersed light. Photodetector is then able to calculate the concentration of particles. The concentration of particles is evaluated using the intensity of the light being received. Figure 1 shows how sensors have been placed in the dust chamber for concentration measurement (Figure 2).

2.2 | Reference instruments

In the present research, three types of reference tools including Thermo Scientific Personal DataRAM pDR-1500 Monitor, DustTrak II Aerosol Monitor, and a PDM3700 Personal Dust Monitor are utilised for evaluation and performance comparison of inexpensive PM sensors. The performance of these three reference instruments was previously compared with each other in our previous work in ref. [25]. Note that Dust-Trak II Aerosol Monitor uses light-scattering method to measure the PM concentration. This tool uses a laser diode as a light source. In addition, Thermo Scientific Personal DataRAM pDR-1500 Monitor is a sensitive nephelometric monitor which uses an LED to generate a beam light for measuring of PM concentration. In contrast to the previous reference tools working based on optical measurement principles, PDM3700 Personal Dust Monitor adopts TEOM to evaluate the concentration of particles [46]. For this experiment, we placed PM sensors along with the reference instruments in the dust



FIGURE 1 Experiment setup in chamber



FIGURE 2 Dust chamber

chamber as can be seen in Figure 1. This setup includes 3 Shinyei PPD42NS sensors, 3 Sharp GP2Y1010AU0F sensors, and 3 Laser SEN0177 sensors. Candidate inexpensive dust sensors and the required hardware and software were previously identified and tested. Our experiments were consisted of placing the low-cost PM dust sensors attached on Arduino boards with data logging capability, into the dust chamber as shown in Figure 1, along with reference instruments including the PDM 3700, Dusttrak DRX, PDR 1500, and personal sampling pump/cyclone to measure respirable dust concentration.

3 | PARTICULATE MATTER SENSORS PARAMETERS

Each PM sensor uses its current-voltage characteristic curve to illustrate its response to an input stimulus. A sensor's characteristic curve may contain some structural errors. Structural errors occur when there are differences between sensor's measured value and the actual value. Thus, calibration process can be utilised to improve the sensor's performance by eliminating the errors in the sensor output. In other words, calibration process is performed to map the sensor's response to an ideal response. Various parameters such as linearity and offset error are evaluated on PM sensors' output values to assist in accomplishing the process of calibration.

3.1 | Linearity of PM sensors

The linearity of PM sensors is defined as the difference between the response of measurements read from PM sensor and the actual concentration which is measured by the reference instruments. We used a relationship between the ideal/expected response curve from DustTrak reference instrument and the actual response curves from PM sensors to determine the linearity of sensor. Linearity is expressed in terms of Percentage of Non-linearity and is shown in the below equation:

$$\text{Nonlinearity (\%)} = \frac{D_{in(max)}}{IN_{fs}} \times 100$$

In which, $D_{in(max)}$ is defined as maximum input deviation and IN_{fs} is defined as the maximum, full-scale input. The non-linearity equation defined above is usually subject to several specific factors related to environment such as humidity, temperature, and vibration. Assuming the post calibrated data has a maximum input scale of 5, the maximum deviations from linear fit are 0.256, 1.848, and 0.967 for Shinyei, Laser and Sharp respectively. Therefore, Shinyei sensor exhibits 5% non-linearity while Laser and Sharp sensors are more non-linear with 36, and 19% of non-linearity respectively.

3.2 | Offset errors of PM sensors

The offset error occurs when the sensor output differs from the actual concentration when the dust concentration is zero. This offset shows up in the characteristic curve despite the slope of this curve matches closely the reference. For example, Shinyei sensor shows a positive offset of 0.110 and Sharp sensor has an offset of 0.246 while the Laser sensor has a small negative offset of -0.023 for ARD. This offset is to some extent removed by the means of calibration equation.

3.3 | Concentration accuracy

Concentration accuracy of sensors was tested by collecting data in a controlled environment. We compared the sensors' calibrated output against the responses obtained from the reference Dust Trak monitor. The maximum difference between the measured value out of calibrated sensor and the reference point was calculated for all measurement points. The precision of each sensor was determined based on the variance of error. For a batch of measurement done in the presence of coal dust the variance for Shinyei sensor was 0.033, whereas Laser had error variance of 0.014, and Sharp sensor had a variance of 0.011 after calibration.

3.4 | Sensitivity

Sensitivity is defined as the smallest amount of variation that is detected by a sensor. For Sharp sensor, as shown in its

datasheet, sensitivity is determined by the amount of output voltage variation when dust density varies by 0.1 mg/m^3 . For sharp sensor, minimum sensitivity, typical sensitivity, and maximum sensitivity are $0.35 \text{ V}/(0.1 \text{ mg/m}^3)$, $0.5 \text{ V}/(0.1 \text{ mg/m}^3)$, and $0.65 \text{ V}/(0.1 \text{ mg/m}^3)$ respectively.

3.5 | Effect of particle size on sensor output

The readout from the sensors depends on the size of used particles. We ran the experiments for two different types of particles: ARD and Coal Dust (CD). Three different sizes of ARD were also used in our experiment. We used ISO 12103-1, A1 Ultrafine Test Dust which has a mean diameter of between 0 and 10 micron size, A2 Fine Test Dust which has a mean diameter of between 0 and 80 micron size and A3 Medium Test Dust which has a mean diameter of between 0 and 80 micron size. The simple linear regression calibration curves were used to adjust the readouts for these two different dust types. They clearly show that the sensors' response is different for these two types of dusts. These results are given in Table 1 and Table 2.

According to these Tables, the calibration process is illustrated in terms of polynomial functions. We used the adjusted R-Squared as the indicator of selecting the order of the polynomial functions (the best model) for each type of sensor. For the Shinyei sensor, the higher polynomial order does not significantly change the residual error and therefore the curve is closer to the linear calibration. This is also clear from coefficients of squared and cubed terms in the fitted polynomial. On the other hand, Laser and Sharp sensors exhibit more non-linearity in their response. The best calibration polynomial models for Sharp and Laser are of degrees two and three, respectively. Statistical tools such as *F*-test [47] are used to choose the best regression model for the calibration procedure. A statistic made of variance ratio, designated as *F*-test, is to test how well the fitted data match the chosen model.

3.6 | Limit of detection

Lower Limit of Detection (LOD) or LOD refers to the lowest quantity that considerably deviates from the signal acquired from blank values. This work calculates the LOD from broadly used Kaiser and Specker method [48]. LOD can be calculated by the following equation.

$$\text{LOD} = 3\gamma/G$$

where γ is defined as standard deviation. Standard deviation calculated at blank conditions which is maintained by filling the dust chamber with air cleaned by high-efficiency particulate air filters. G is also defined as the fitted line slope achieved from linearity of experiments. For example, the LOD for Shinyei, Laser and Sharp are $3 \times 0.181/0.668 = 0.812 \text{ mg/m}^3$, $3 \times 0.0116/3.379 = 0.103 \text{ mg/m}^3$, and $3 \times 0.103/2.982 = 0.110 \text{ mg/m}^3$ respectively.

4 | RESULTS AND DISCUSSIONS

4.1 | Data preparation and data collection

In this study, three different types of off-the-shelf PM sensors (Shinyei PPD42NS, Sharp GP2Y1010AU0F, Laser SEN0177) have been selected for evaluation. The output of each sensor is different. The output of the PPD4 sensor is modulated pulses. PPD4 sensor measures the PM level by counting the Low Pulse Occupancy time in given time unit. The PPD4 sensors can detect particles with a size range of $0.5\text{--}2.5 \mu\text{m}$. The output of SEN0 is digital pulses. The SEN0 computes the number of suspended PM in a unit volume of air within a range of between 0.3 and $10 \mu\text{m}$. Therefore, it can identify particles ranging from 0.3 to $10 \mu\text{m}$. The output of the GP2Y sensor is an analog voltage. It then uses a mathematical formula to convert the analog voltage into dust

TABLE 1 Calibration equations and best calibration models for different types of particulate matter (PM) sensors when Arizona Road Dust (ARD) was used as the main source of producing dust particles

Equation	R-Sq (adj) (%)	DF	F-Test	p-value
DustTrak = $0.1106 + 0.6686 \text{ Shinyei} - 0.06616 (\text{Shinyei})^2 + 0.007649 (\text{Shinyei})^3$	84.5	3.1073	1954.37	0.0001
DustTrak = $-0.02383 + 3.379 \text{ laser} + 5.247 (\text{Laser})^2 - 6.780 (\text{Laser})^3$	79.3	3.1073	1375.20	0.0001
DustTrak = $0.2468 + 2.982 \text{ sharp} + 2.927 (\text{Sharp})^2$	85.1	2.1074	3073.70	0.0001

TABLE 2 Calibration equations and best models for different types of particulate matter (PM) sensors when Coal Dust was used as the main source of producing dust particles

Equation	R-Sq (adj) (%)	DF	F-Test	p-value
Dust trak = $0.1529 + 2.046 \text{ Shinyei} - 0.6870 (\text{Shinyei})^2$	83.9	2.357	938.82	0.0001
Dust trak = $0.07471 - 0.1288 \text{ laser} + 9.269 (\text{Laser})^2 - 8.047 (\text{Laser})^3$	95.3	3.356	2454.22	0.0001
Dust trak = $0.1561 + 1.820 \text{ sharp} + 8.325 (\text{Sharp})^2 - 11.23 (\text{Sharp})^3$	93.9	3.356	1842.83	0.0001

density values. The GP2Y sensor can identify particle sizes of approximately less than $0.8\ \mu\text{m}$. A total of nine low-cost sensors have been used for data collection in a controlled environment. Sensors were connected to the battery-powered Arduino boards (Arduino Uno REV3) which are equipped with data logging shields to be able to collect data. To carry out the experiment, low-cost sensor units along with standardised instruments such as PDR 1500, PDM 3700, DustTrak DRX, and personal sampling pump/cyclone were placed into the dust chamber as shown in Figure 1. PDM 3700 and personal sampling pump both have been considered as two of the main reference instruments for measurement of the respirable dust. Both devices yield gravimetric estimations of the concentration of respirable dust. Moreover, the PDM 3700 instrument records a 15-min of time-weighted average (TWA) for measurement of the respirable dust concentration to assist in comparing the dust sensor output. For this research, the performance of all three reference instruments (PDM 3700, PDR 1500, and DustTrak DRX) have been compared with each other and shown in [25]. Figure 3 in ref. [25] indicates the comparison of the 15-min TWA respirable dust concentration for PDM 3700 along with corrected PDR 1500 and corrected DustTrak DRX. Data was collected in a controlled environment from the three different types of PM sensors along with the reference instruments over a course of 4 days. The type and size of particles used in each experiment were varied. We utilised ARD in the first three experiments, but each time we used varied particle sizes. We utilised A1 Ultrafine Test Dust on the first day, A2 Fine Test Dust on the second day, and A3 Medium Test Dust on the third day. We utilised coal dust on the final day of the experiment. To further evaluate the properties of selected off-the-shelf PM sensors, their responses were plotted against the response we collected from the DustTrak reference instrument as shown in Figures 3 and 4.

In Figures 3 and 4 as can be seen, the PM concentrations for laser and sharp sensors fluctuates from initial point of $0\ \text{mg}/\text{m}^3$ to around $0.5\ \text{mg}/\text{m}^3$ in all figures, while the PM concentrations for Shinyei sensors in Figure 3 where we used A1 Ultrafine Arizona Test Dust fluctuates from $0\ \text{mg}/\text{m}^3$ to around $5.5\ \text{mg}/\text{m}^3$, in Figure 3 on the right, where we used A2 Fine Arizona Test Dust fluctuates from $0\ \text{mg}/\text{m}^3$ to around $4.5\ \text{mg}/\text{m}^3$, in Figure 4 on the left, where we used A3 Medium Arizona Test Dust fluctuates from $0\ \text{mg}/\text{m}^3$ to around $4.5\ \text{mg}/\text{m}^3$ and in Figure 4 on the right, where we used different type of dust (coal dust) fluctuates from $0\ \text{mg}/\text{m}^3$ to around $1.6\ \text{mg}/\text{m}^3$. Thus, the output responses of Shinyei sensors are shown to be more sensitive to different size/type of particles compared to laser and sharp sensors. In addition, as can be seen, laser and sharp sensors' output responses are correlated, underestimate the concentration of particles, and less sensitive to finer dust particle size. The sensors' output signals are expressed in terms of voltages. Sensors' output signals are then converted into dust density values based on their manufacturers' guidelines. As can be seen in Figures 4 and 5, the PM sensors' output responses are not completely matched with the output response of the reference instrument. This incongruity however can be compensated with an appropriate calibration technique.

4.2 | Calibration using sensor fusion

In this section, we aim to better improve the calibration of low-cost PM sensors using a combination of different sensor types. To that end, eight different ML techniques have been compared against each other. These ML techniques include RF, Gradient Boosting, Multilayer perceptron (MLP), Support Vector Machines (SVM), Ridge Regression, Lasso Regression, Elastic Net, and Stacking Regression.

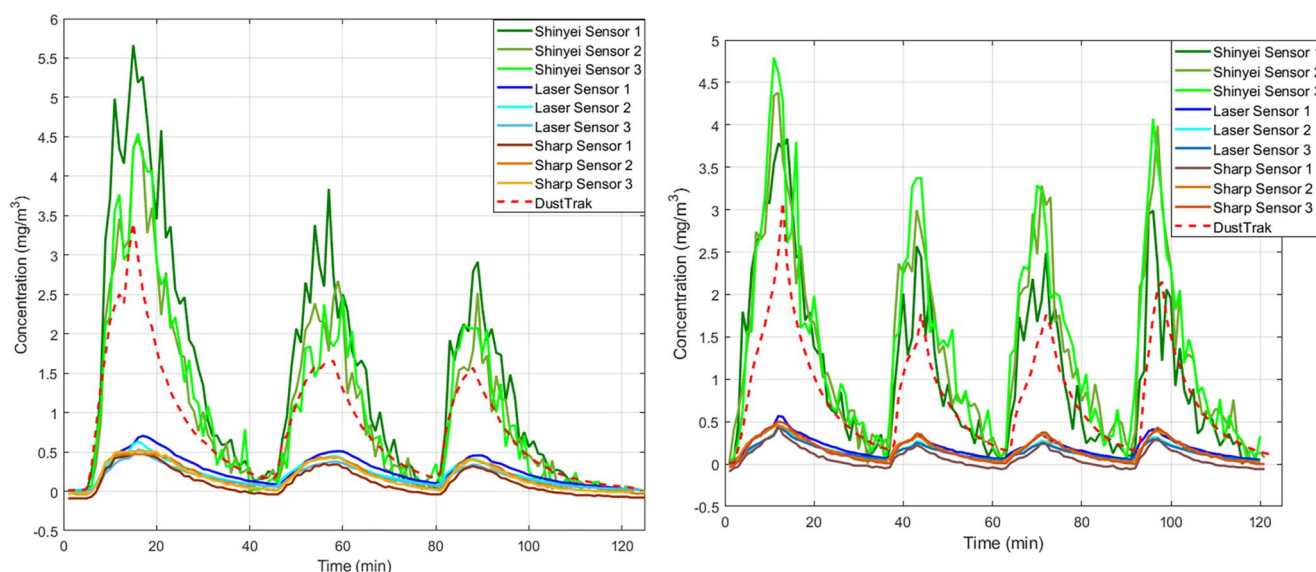


FIGURE 3 Uncalibrated raw data—On the left: with Arizona Test Dust- ISO 12103-1-A1 Ultrafine, One the right: with Arizona Test Dust- ISO 12103-1-A2 Fine.

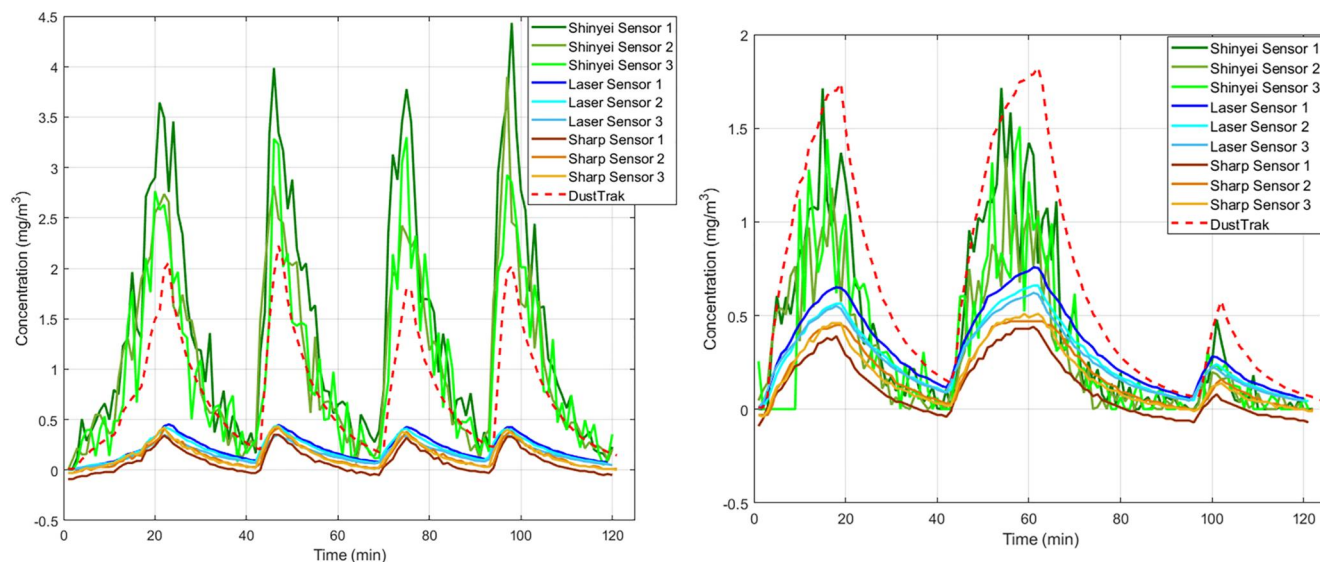


FIGURE 4 Uncalibrated raw data—On the left: Arizona Test Dust- ISO 12103-1-A3 Medium, One the right: with Coal Dust.

(a) Random forest

Random Forest algorithm is one of the widely used ML algorithms for classification and regression. In this method, the aggregate of the response of multiple sensors provides a better calibration compared to the best individual sensor. The RF model works based on the strategy of providing a prediction based on individual prediction collected from each sensor, called means ensemble technique [49]. The backbone of RF is the decision tree [50] where each of the subtrees partitions the data in the regression trees based on if-else rules [51]. The Gini index or the Entropy index are used for splitting or partitioning the data in the regression trees.

(b) Gradient Boosting

Gradient Boosting is another method of increasing the accuracy of calibration from different sensor types with subpar accuracy. The same strategy of RF is employed in gradient boosting as well, however, the training is different. In RF the training is done for each individual sensor, but in gradient boosting the k^{th} sensor is trained from the first $k-1$ sensors responses and update the residual based on difference between the calibrated value and the dust density (real value) [52].

(c) Multilayer perceptron

Multilayer perceptron [53] is a type of feed-forward neural network with one or more interconnected layers between the input and the output. Feed-forward means that data flows in one direction from input to output to transfer the representation towards the last layer. To train the feed-forward networks, backpropagation learning procedure is the most common used method [54]. MLP is a supervised learning model where the output $f(x)$ can be compared with a target value of Y through a loss function to minimise the amount of

error and the difference between the output and target. In our calibration process, we have chosen the dust density as the target value. The calibration inaccuracy, as the loss function, is minimised by the MLP.

(d) Support Vector Machines

Support Vector Machines is one of the supervised algorithms that we used here. The SVM models the input values in a finite dimensional feature space by representing the individual sensor output on different dimensions. Using a kernel function instead of the linear boundary helps with non-linear fitting [55]. The representation of the sensors' output is mapped to the final calibrated value.

(e) Penalised regression models

The standard linear regression model (or the ordinary least square method) performs poorly in situations where there is a high complexity in a multivariate data set. An alternative method named regularised regression (or penalised regression) creates a penalised linear regression model by adding a constraint called regularisation. This penalty forces the model to limit the growth of coefficient values. The added constraint leads to small coefficient values for the less contributing variables. Generally, there are three main strategies to penalise the ordinary linear regression:

• Ridge Regression

The goal of penalisation is to decrease the model complexity, or the number of sensors which have no contribution to the calibration. Instead of completely removing an individual sensor from the data set, its coefficient is chosen close to zero if there is low correlation between its response values and dust density. The basic strategy for Ridge regression model [56] for achieving

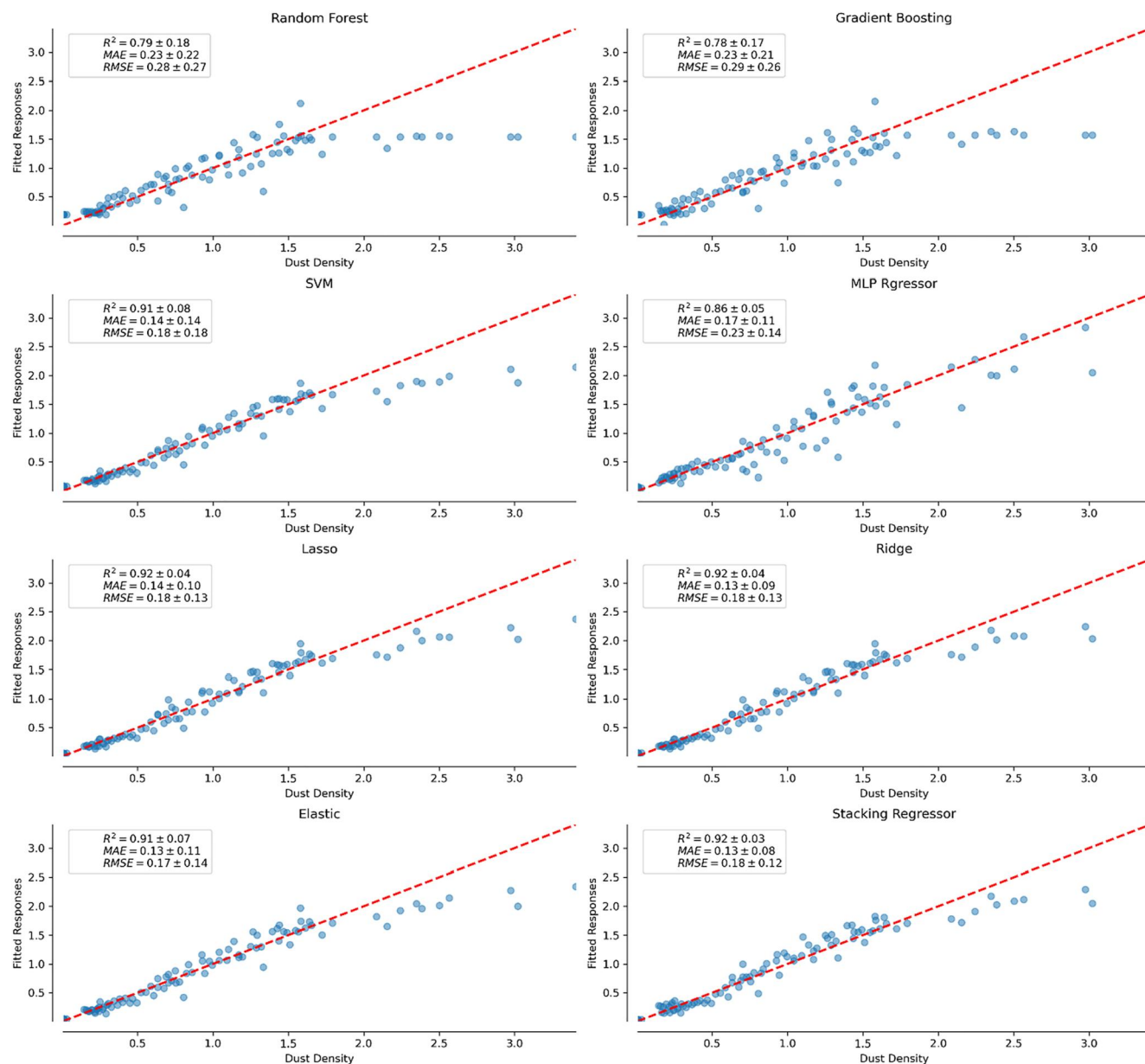


FIGURE 5 Machine Learning (ML) methods used for sensor calibration (Arizona Road Dust (ARD)).

the penalised coefficients is attained by adding a penalty term which is the sum of the squared coefficients.

- Lasso Regression

Lasso (Least Absolute Shrinkage and Selection Operator), is based on the same concept as ridge regression [57]. This model adds a penalty term in the loss function to penalise the non-zero coefficients, but unlike the Ridge regression, the Lasso penalises with the sum of the absolute values.

- Elastic Net

The drawback in the Lasso model is that the variable selection can be dependent on the data and thus unstable. Elastic

Net combines the penalties from the Ridge regression and the Lasso regression to take the advantage of both models.

Figures 5 and 6 depict the fitting the sensor output with different ML techniques for Arizona road and coal dust data sets respectively. The model combines the response of 9 different sensors responses to increase the accuracy of dust density readouts. In the subfigures the x-axis is the measured dust density based on the DustTrak (reference) instrument, and y-axis is the output of regression model from sensors response measurements collected in the experiment. To evaluate the fit three different parameters are shown. (1) Root mean square error (RMSE) finds the average of squared errors between predicted value and the real dust density. (2) Mean absolute error (MAE) is alternatively illustrates the average of absolute error between the pair of predicted output and dust

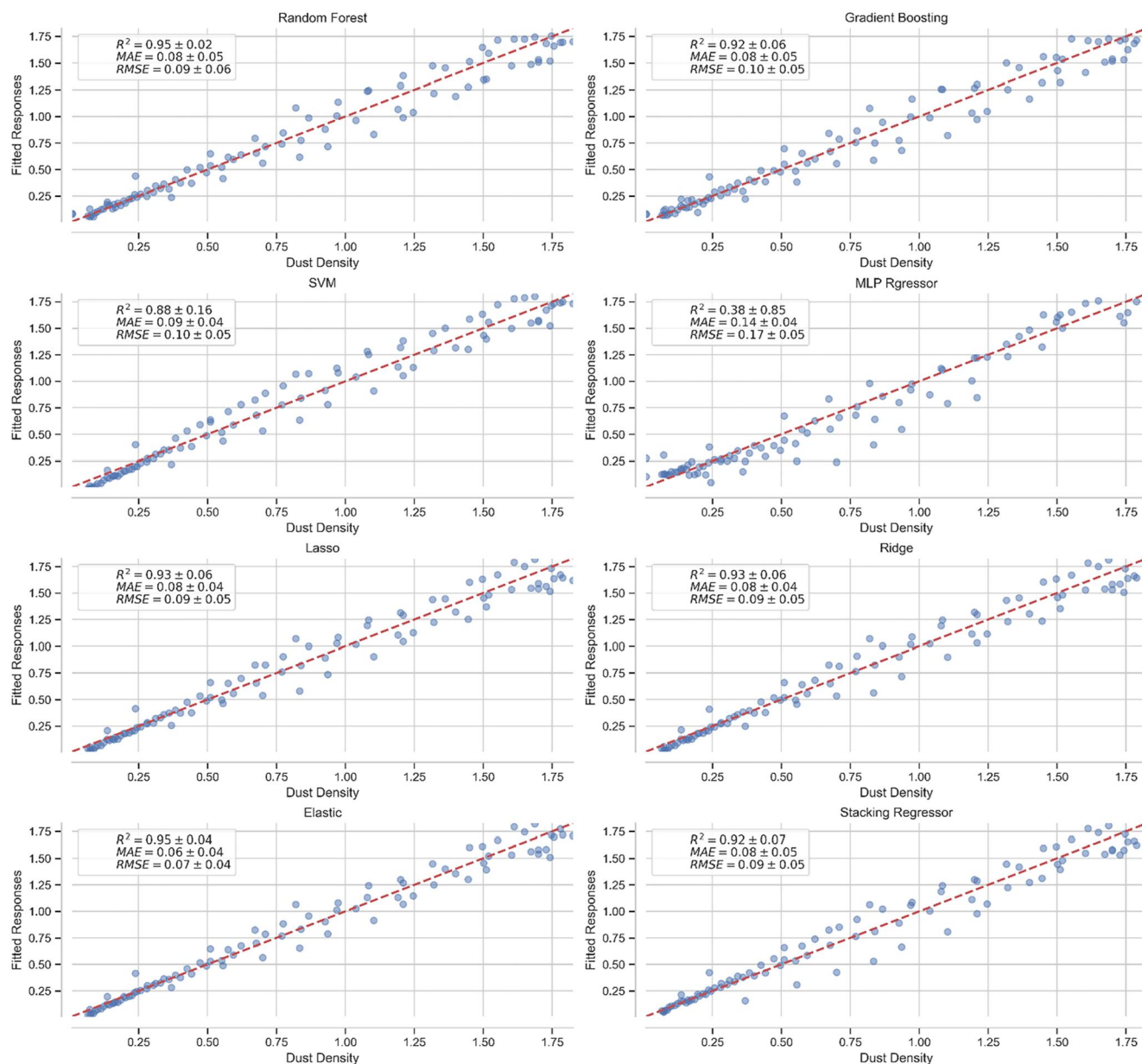


FIGURE 6 Machine Learning (ML) methods used for sensor calibration (Coal Dust).

density. (3) R squared as ratio of variance of technique's output values that is predicted from the dust density values. All the models show reliable predictions of dust density, but for the experiment result used the SVM method has the highest R squared value. This complies with the lower RMSE value for SVM method compared to the other methods illustrated. Therefore, SVM is the most promising technique in the evaluation of sensor responses against dust density measurements. The model trained was used to calibrate the sensors' output for measurements. The output of different methods was used in the previous modelling for the calibration of sensors. The calibrated response from the nine sensors is presented in Figures 7 and 8 against the dust density measure by the reference data. The results in these figures show a good fit at different points of the experiment in the chamber where the

Arizona road and coal dust densities are adjusted over the range of 0–3.5 mg/m^3 and –0.3–2.5 mg/m^3 respectively.

It must be noted that, for data analysis using ML models, new features have not been created. All the features are the same as raw data. In fact, there have been no manipulation, transformation, or any other feature engineering techniques done on the dataset for developing ML models. Eventually, scikit learn library in Python has been used to implement all the ML models. The dataset that has been used for the data analysis had multiple samples. This dataset was contained a table with 121 rows and 10 columns. Each row contained nine columns indicating output information of the low-cost PM sensors while the 10th column represented the output of the reference instrument. The size of the dataset was sufficient in order to achieve an accurate estimate of the model

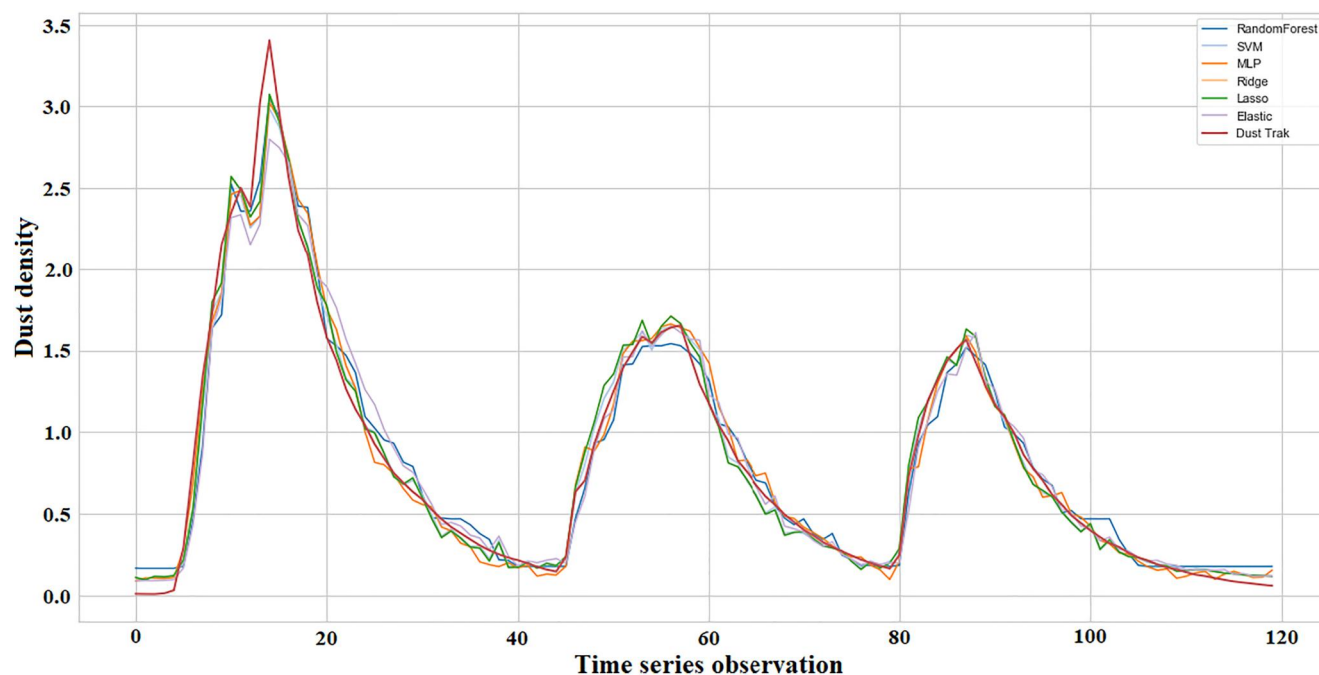


FIGURE 7 Calibrated data using Machine Learning (ML) techniques for Arizona Road Dust (ARD).

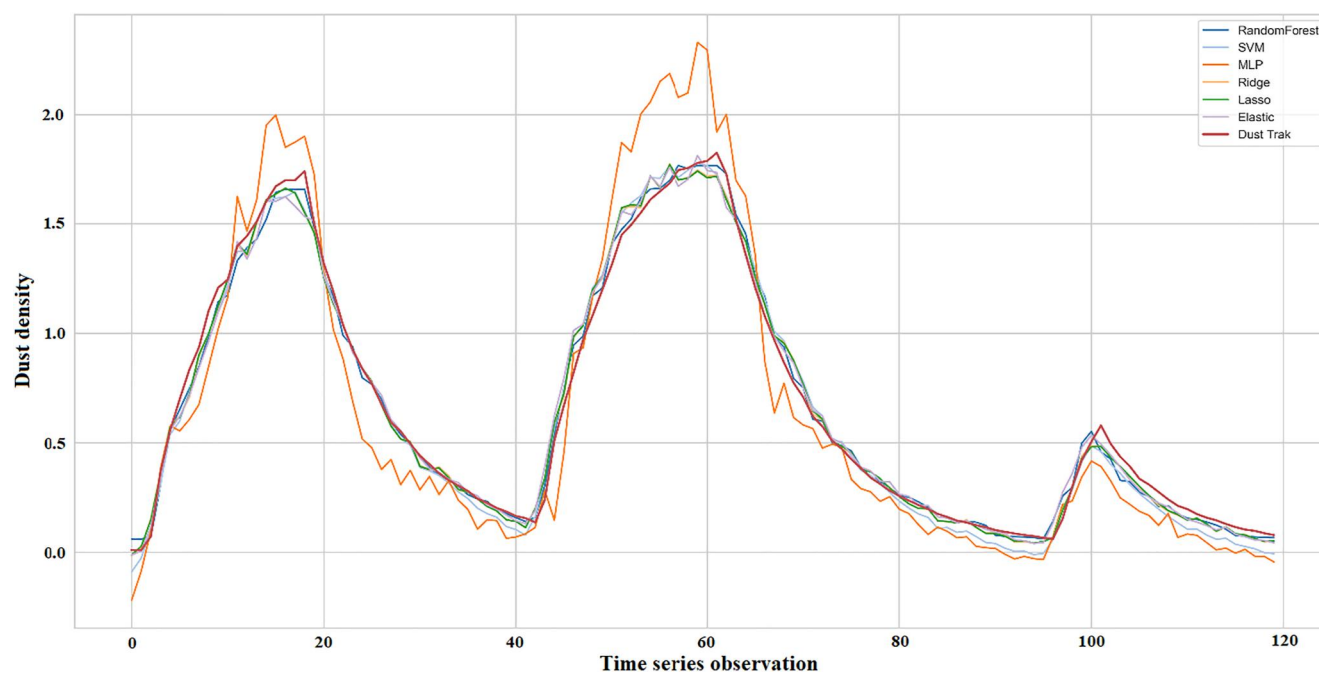


FIGURE 8 Calibrated data using Machine Learning (ML) techniques for Coal Dust.

performance. Samples have been formatted with an implied decimal point and up to seven decimal digits have been indicated. In addition, the dataset was normalised so that the results would not be skewed toward the majority of the data. In all these experiments for ML-based models, we split the dataset into train, validation, and test with different percentage of 60%, 20%, and 20% respectively and the results reported in this work are based on unseen dataset, which means the test

dataset. As a matter of fact, all ML models need to be tuned due to having different hyper-parameters to generalise the model and control the overfitting problem. To do so, the grid-search technique is used to find the optimal hyper-parameter for each model based on a 5-fold cross validation technique. In the Table 3 you can see the range of values selected for optimisation and also the best value picked by grid-search technique for each model.

TABLE 3 Machine Learning (ML) models with all related parameters used in this research

Algorithm	Hyperparameter	Range of value for test	Optimal value
Random forest	Number of estimators (trees)	{20, 40, 60, 80, 100, 120, 140, 160, 180, 200}	80
	Min_sample_split	{2, 5, 8, 10, 12, 14}	8
	Min_sample_split	{1, 2, 4, 6, 8, 10}	6
	Max depth of tree	{2, 3, 5, 6, 8, 10, none}	5
Gradient boosting	Max_leaf_node	{2, 4, 6, 8, 10, 12, 14, 16, 18, 20}	10
	Min_sample_leaf	{1, 2, 4, 6, 8, 10}	4
	L2_regularisation	{1, 10^{-2} , 10^{-4} , 10^{-6} , 10^{-8} , 10^{-10} }	10^{-4}
SVM	Kernel	Linear	Linear
	C (regularisation term)	{1, 10, 100, 1000}	1
MLP	Hidden layer number	{1, 2, 3}	1
	Range of hidden layer neurons	{1–100}	50
	Activation function	{Logistic, tanh, ReLU}	ReLU
	Optimisation technique	{sgd, Adam}	Adam
	Alpha	{0, 0.001, 0.0005, 0.00005}	0.001
	Max iteration	500	500
Ridge regression	L_2 penalty	{1, 10^{-2} , 10^{-4} , 10^{-6} , 10^{-8} , 10^{-10} }	10^{-2}
Lasso regression	L_1 penalty	{1, 10^{-2} , 10^{-4} , 10^{-6} , 10^{-8} , 10^{-10} }	10^{-4}
Elastic net	L_2 penalty	{1, 10^{-2} , 10^{-4} , 10^{-6} , 10^{-8} , 10^{-10} }	10^{-2}
	L_1 penalty	{1, 10^{-2} , 10^{-4} , 10^{-6} , 10^{-8} , 10^{-10} }	10^{-4}

The common way to tune the hyperparameter and select the optimal value for them in more ML-like approaches is to perform cross-validation and select the value that minimises the cross-validated sum of squared residuals (or some other measure). As it can be seen in Table 3, to choose the value for each hyperparameter through cross-validation, a set of values for each of them is chosen to test. Then, the dataset is split into k folds, and each time use the $k-1$ fold for training the model and use the rest of data for testing the model and repeat this strategy for k times of k folds. The optimal value for each hyperparameter is selected based on minimisation of the sum of squared residuals. In this work, cross-validation with 5 folds has been applied to detect the optimal value for each of hyperparameters.

Note that, for all ML models, standardising the input data is important. If the predictors are not standardised, their standard deviations are not all equal to one. This situation can make a bias in training process of ML and has negative impact on final prediction because the unstandardised coefficients are then weighted by the inverse of the standard deviations of their corresponding predictors. For example, for regression models, this scaling ensure that the penalty term penalises each coefficient equally. In this work, the input dataset is scaled to the standard deviation of 1 and mean value of zero.

Random Forest and gradient boosting models are ensemble models to empower the weaker models such as decision tree by combining the results from them. In this work, for RF and gradient boosting, between 20 and 200 trees has

been tested in order to find the optimal number of trees as a predictor. There are several ways to control the overfitting problem in RF and gradient boosting models, such as limit the depth of trees for splitting the input samples and find the optimal number of samples for each node and each leaf for each tree. In this work, we set different values for each of these hyperparameters for RF and gradient boosting models by using the cross-validation technique which end up the values presented in Table 3 for RF and gradient boosting models.

SVM is kernel based model with three main kernel methods, linear, multinomial, and RBF kernels. In this work, linear kernel method as parametric technique has been used to provide the coefficients of different input features. Like all ML models, SVM is prone to falling in overfitting problem, which regularisation term (C) has been added to loss function to ameliorate the overfitting. In SVM, model tries to find a hyperplane with largest minimum margin and minimise the mispredicting or misclassification rate. The C value tries to find the higher value for margin with lowest mis-predicting. In this work, different value between 1 and 1000 has been tested based on cross-validation technique and the optimal value for linear SVM is 10 for input data in this work.

For MLP model, different combination of number hidden layer (1, 2, and 3), number of neuron for each layer (between 1 and 100), different optimisation and activation techniques have been tested. The best combination and fit model for MLP based on input data is obtained by having only one layer with 50 neurons, the activation function is ReLU with Adam

TABLE 4 The comparison of the eight different Machine Learning (ML) models with each other (Arizona Road Dust (ARD))

Algorithm	R^2	MAE	RMSE
Random forest	0.79 ± 0.18	0.23 ± 0.22	0.28 ± 0.27
Gradient boosting	0.78 ± 0.17	0.23 ± 0.21	0.29 ± 0.26
SVM	0.91 ± 0.08	0.14 ± 0.14	0.18 ± 0.18
MLP	0.86 ± 0.05	0.17 ± 0.11	0.23 ± 0.14
Lasso regression	0.92 ± 0.04	0.14 ± 0.10	0.18 ± 0.13
Ridge regression	0.92 ± 0.04	0.13 ± 0.05	0.18 ± 0.13
Elastic net	0.91 ± 0.07	0.13 ± 0.11	0.17 ± 0.14
Stacking regression	0.92 ± 0.03	0.13 ± 0.08	0.18 ± 0.12

TABLE 5 The comparison of eight different Machine Learning (ML) models with each other (Coal Dust)

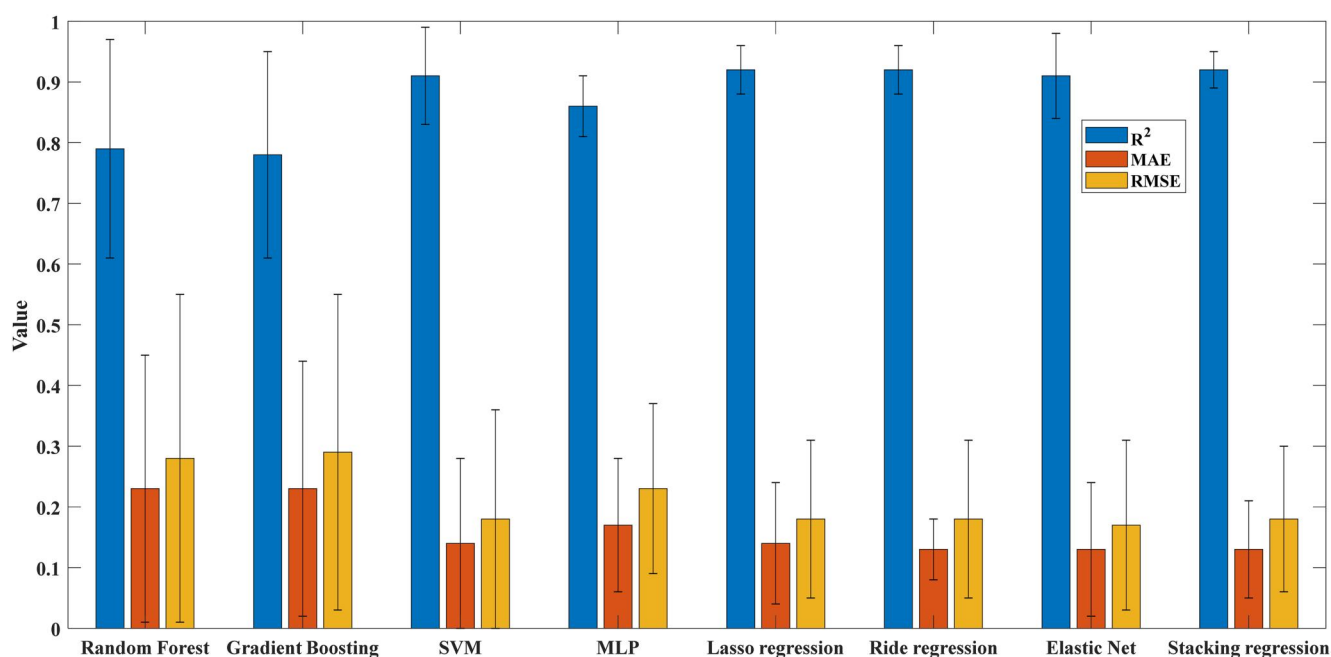
Algorithm	R^2	MAE	RMSE
Random forest	0.95 ± 0.02	0.08 ± 0.05	0.09 ± 0.06
Gradient boosting	0.92 ± 0.06	0.08 ± 0.05	0.10 ± 0.05
SVM	0.88 ± 0.16	0.09 ± 0.04	0.10 ± 0.05
MLP	0.38 ± 0.85	0.14 ± 0.04	0.17 ± 0.05
Lasso regression	0.93 ± 0.06	0.08 ± 0.04	0.09 ± 0.05
Ridge regression	0.93 ± 0.06	0.08 ± 0.04	0.09 ± 0.05
Elastic net	0.95 ± 0.04	0.06 ± 0.04	0.07 ± 0.04
Stacking regression	0.92 ± 0.07	0.08 ± 0.05	0.09 ± 0.05

optimisation technique. In order to avoid overfitting problem, different value for alpha have been tested where the best value is 0.001. For all three regularised regression models, different values have been tested for L1 and L2 penalty term which end up to 10^{-2} for L1 and 10^{-4} for L2.

A comparison of the 8 ML models in terms of the R^2 , MAE, and RMSE values to select the best algorithm is presented for the ARD and one type of coal dust in Tables 4 and 5 respectively. In addition, the performance comparison of the ML models can be shown in Figures 9 and 10. Finally, evaluation and comparison of 8 different ML techniques exhibit that for ARD, Lasso and Ridge and attacking Regressions had higher adjusted R-squared and therefore better performance, while for Coal Dust, RF and Elastic Net had higher adjusted R-squared and therefore better performance in comparison to others.

In the present study, the 'scikit learn' library in Python is employed to implement all the ML models. Besides, the sensors and modules for the experimental setup are listed in Table 6. It is worth mentioning that the specifications of software, hardware, and other tools used in this research along with their version are detailed in this table.

It is worth mentioning that evaluation of the performance of the low-cost PM sensors using different ML techniques already exist in the literature. As a result, we have explained relevant studies and what innovation have been offered in light of our work. It is worth mentioning that as the dataset has been collected by the authors and cannot be found elsewhere, the numerical comparison with the previous studies is not possible. Hence, we have tried our best to explain what the novelties of the current study are as compared to the earlier research works.

**FIGURE 9** The comparison of the eight different Machine Learning (ML) models with each other (Arizona Road Dust (ARD)).

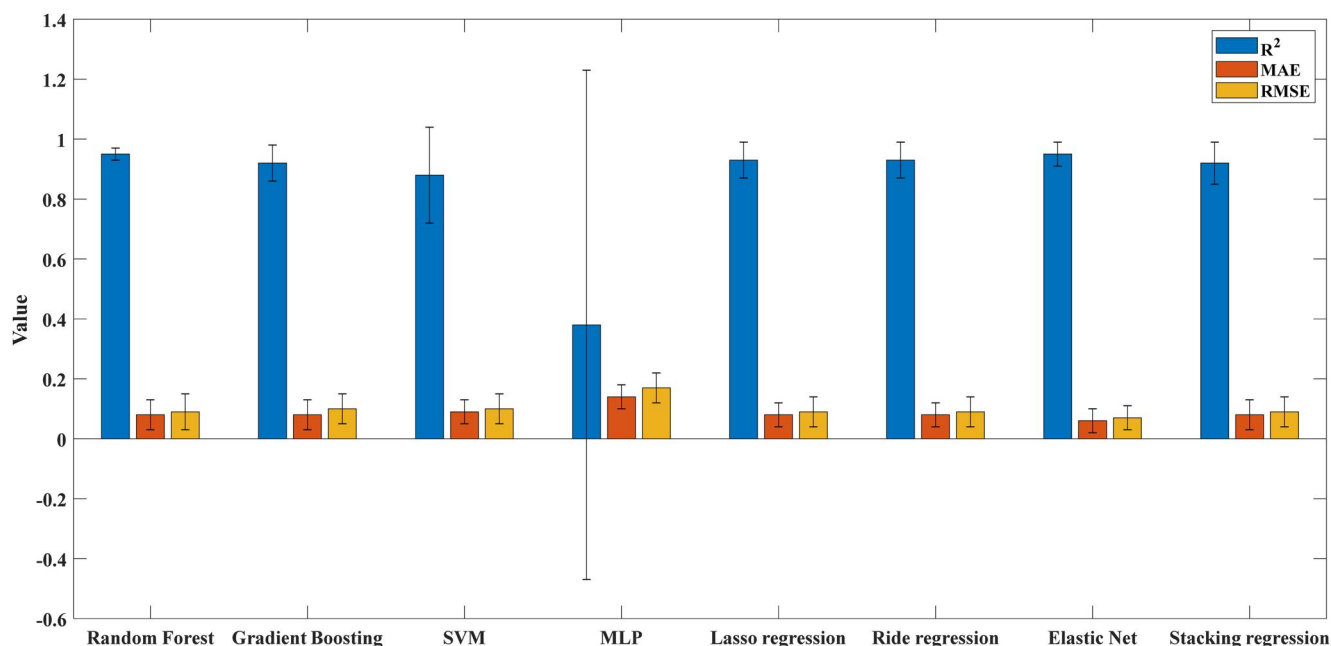


FIGURE 10 The comparison of eight different Machine Learning (ML) models with each other (Coal Dust).

Hardware	Version
Arduino boards	Uno REV3
Laptop	—
Low-cost PM sensor no. 1	Shinyei PPD42NS
Low-cost PM sensor no. 2	Sharp GP2Y1010AU0F
Low-cost PM sensor no. 3	Laser SEN0177
Aerosol monitor no. 1	Thermo Scientific personal DataRAM pDR 1500
Aerosol monitor no. 2	DustTrak DRX
Aerosol monitor no. 3	PDM3700
Aerosol monitor no. 4	Personal pump

TABLE 6 Specification of the experimental set up

5 | CONCLUSION

In this paper, we evaluated the accuracy of each of the three different types of inexpensive, light-scattering-based PM sensors (Shinyei PPD42NS, Sharp GP2Y1010AU0F, and Laser SEN0177) using ML algorithms. To this end, PM sensors along with standardised air quality monitoring devices were exposed to different concentrations of respirable dust. The response of inexpensive PM sensors with the standardised instruments were evaluated. To calibrate each of the off-the-shelf PM sensors and to combine their data to obtain better accuracy, several ML techniques were adopted. The results revealed that, while these sensors may be utilised as a cheaper alternative to more expensive options, they have a different level of accuracy. The methods outlined here can be used to determine if the sensor meets the accuracy requirements of the intended application. The use of such

sensors will allow for the deployment of a large network in the field at various places. After collecting data wirelessly and storing it in the cloud for a period, a more accurate image of the environment may be obtained. The available new techniques in ML and data fusion are ripened to make sense of this large amount of data. These techniques exhibit promising results to increase the accuracy of response from multiple sensors.

AUTHOR CONTRIBUTIONS

Mohammad Ghamari: Conceptualisation; Data curation; Formal analysis; Methodology; Validation; Visualisation; Writing – original draft; Writing – review & editing; Supervision. **Hamid Kamangir:** Formal analysis. **Keyvan Arezoo:** Writing – original draft; Writing – review & editing. **Khalil Alipour:** Writing – original draft; Writing – review & editing; Supervision.

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CONFLICT OF INTEREST

I confirm that none of the authors have a conflict of interest to disclose.

DATA AVAILABILITY STATEMENT

Data that support the findings of this study are available on request from the corresponding author. Data are not publicly available due to privacy or ethical restrictions.

ORCID

Mohammad Ghamari  <https://orcid.org/0000-0002-7752-4270>

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